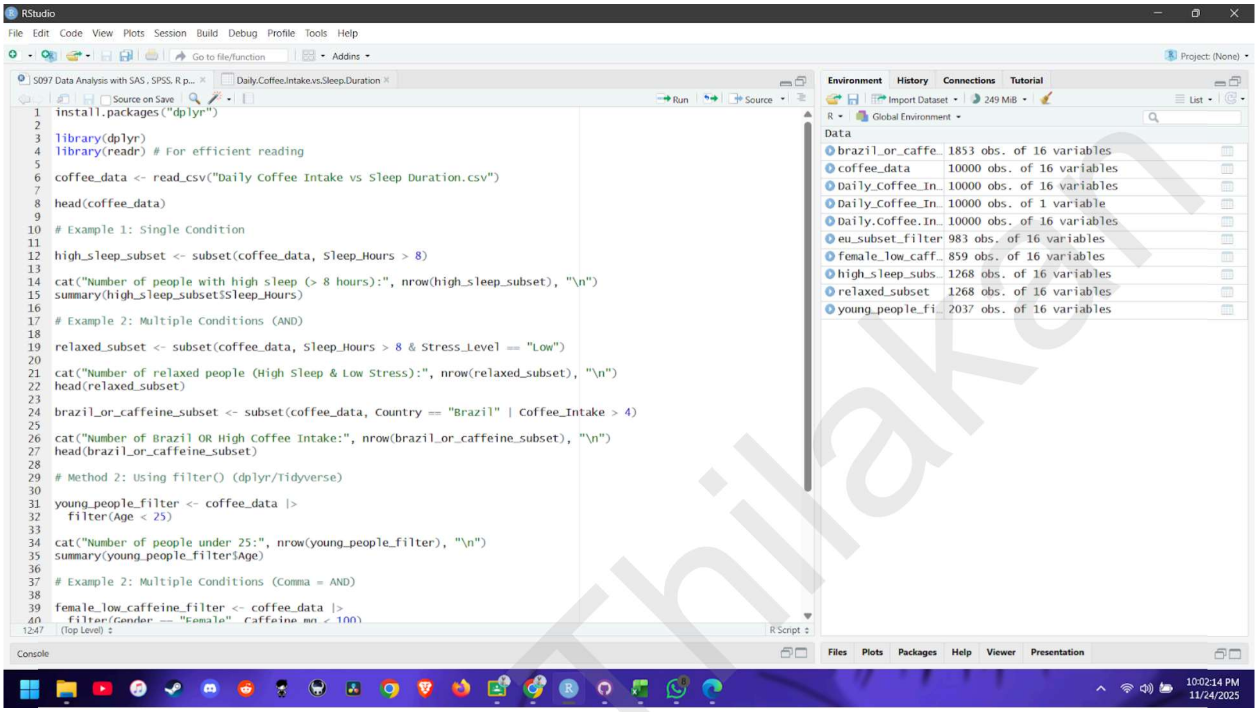
SHETH LUJ AND SIR MV COLLEGE
Subject: Data Analysis with SAS / SPSS /R

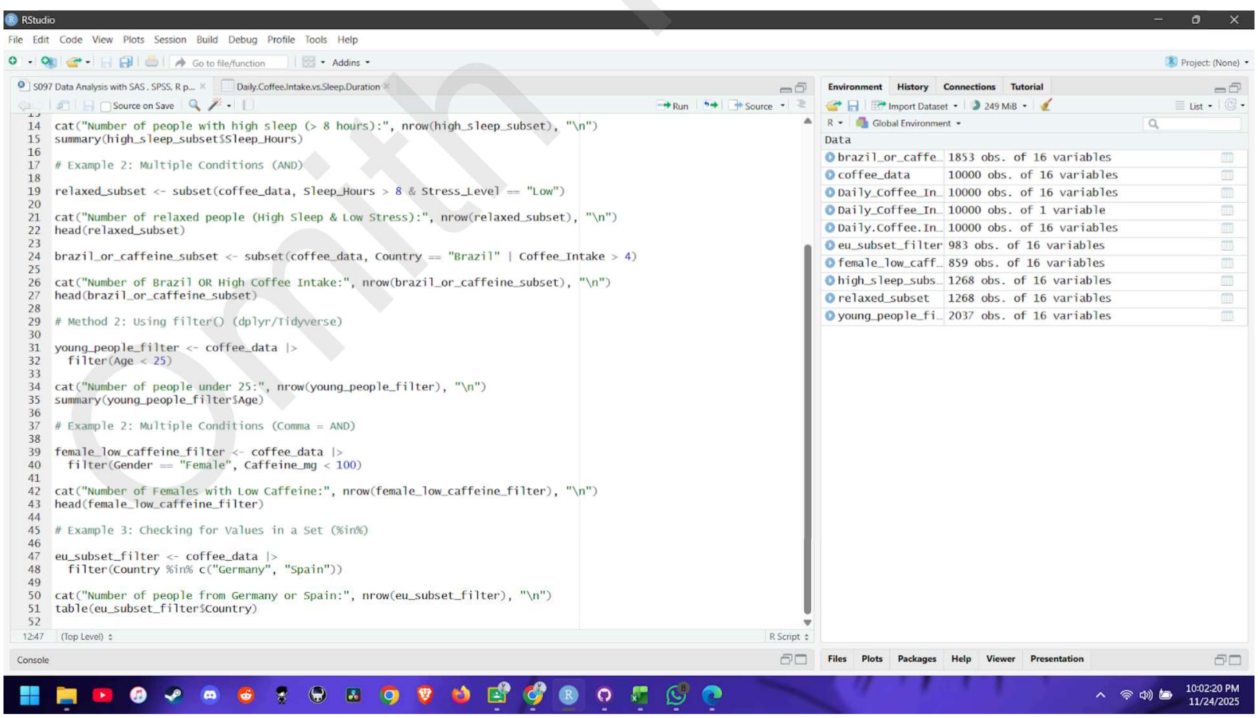
Practical No: 4

Aim: Applying conditional filters subset() or filter() in R.

Code:



```r
install.packages("dplyr")

library(dplyr)
library(readr) # For efficient reading

coffee_data <- read_csv("Daily Coffee Intake vs Sleep Duration.csv")

head(coffee_data)

# Example 1: Single Condition

high_sleep_subset <- subset(coffee_data, Sleep_Hours > 8)

cat("Number of people with high sleep (> 8 hours):", nrow(high_sleep_subset), "\n")
summary(high_sleep_subset$Sleep_Hours)

# Example 2: Multiple Conditions (AND)

relaxed_subset <- subset(coffee_data, Sleep_Hours > 8 & Stress_Level == "Low")

cat("Number of relaxed people (High Sleep & Low Stress):", nrow(relaxed_subset), "\n")
head(relaxed_subset)

brazil_or_caffeine_subset <- subset(coffee_data, Country == "Brazil" | Coffee_Intake > 4)

cat("Number of Brazil OR High Coffee Intake:", nrow(brazil_or_caffeine_subset), "\n")
head(brazil_or_caffeine_subset)

# Method 2: Using filter() (dplyr/Tidyverse)

young_people_filter <- coffee_data |>
  filter(Age < 25)

cat("Number of people under 25:", nrow(young_people_filter), "\n")
summary(young_people_filter$Age)

# Example 2: Multiple Conditions (Comma = AND)

female_low_caffeine_filter <- coffee_data |>
  filter(Gender == "Female", Caffeine_mg < 100)
```



```r
cat("Number of people with high sleep (> 8 hours):", nrow(high_sleep_subset), "\n")
summary(high_sleep_subset$Sleep_Hours)

# Example 2: Multiple Conditions (AND)

relaxed_subset <- subset(coffee_data, Sleep_Hours > 8 & Stress_Level == "Low")

cat("Number of relaxed people (High Sleep & Low Stress):", nrow(relaxed_subset), "\n")
head(relaxed_subset)

brazil_or_caffeine_subset <- subset(coffee_data, Country == "Brazil" | Coffee_Intake > 4)

cat("Number of Brazil OR High Coffee Intake:", nrow(brazil_or_caffeine_subset), "\n")
head(brazil_or_caffeine_subset)

# Method 2: Using filter() (dplyr/Tidyverse)

young_people_filter <- coffee_data |>
  filter(Age < 25)

cat("Number of people under 25:", nrow(young_people_filter), "\n")
summary(young_people_filter$Age)

# Example 2: Multiple Conditions (Comma = AND)

female_low_caffeine_filter <- coffee_data |>
  filter(Gender == "Female", Caffeine_mg < 100)

cat("Number of Females with Low Caffeine:", nrow(female_low_caffeine_filter), "\n")
head(female_low_caffeine_filter)

# Example 3: Checking for Values in a Set (%in%)

eu_subset_filter <- coffee_data |>
  filter(Country %in% c("Germany", "Spain"))

cat("Number of people from Germany or Spain:", nrow(eu_subset_filter), "\n")
table(eu_subset_filter$Country)
```

Output:

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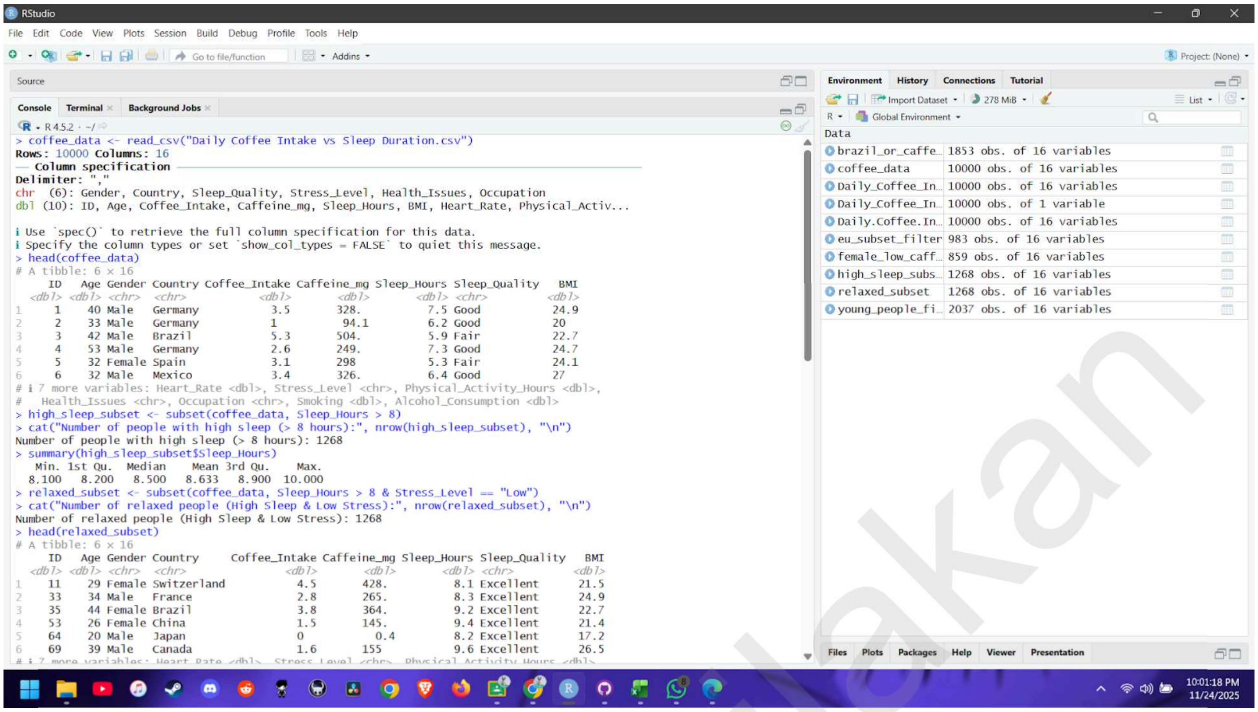
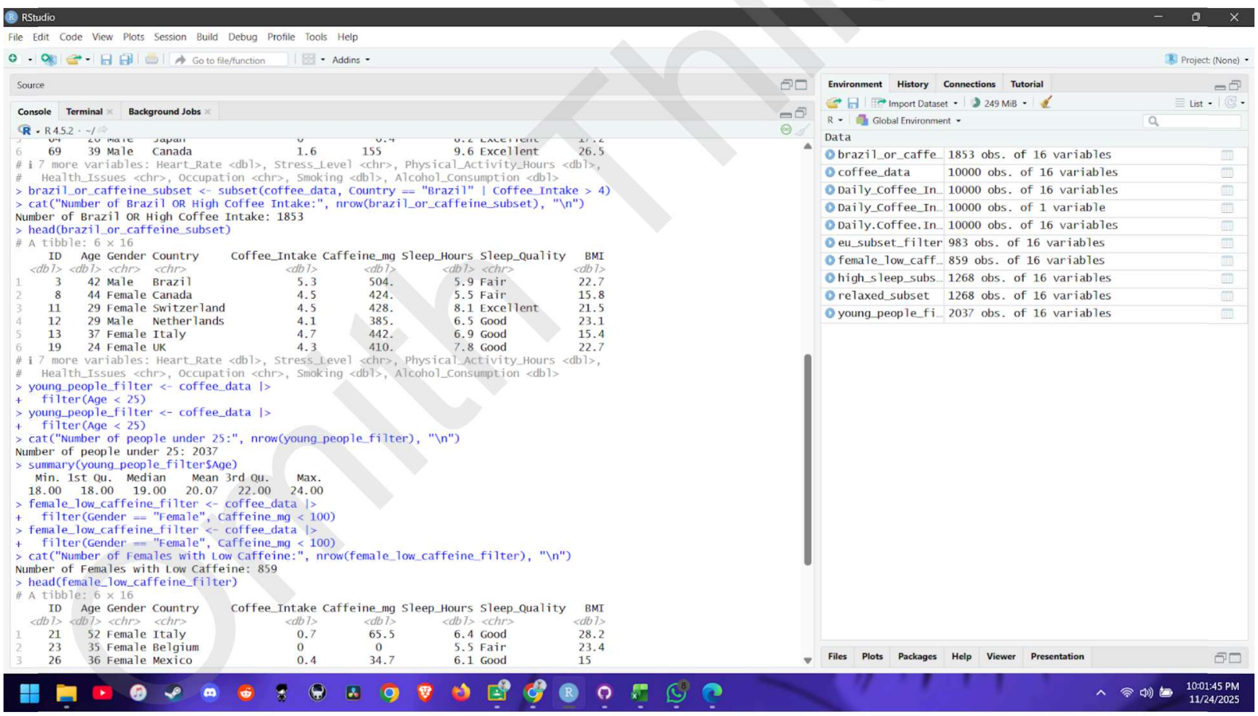
S097



```
> coffee_data <- read_csv("Daily Coffee Intake vs Sleep Duration.csv")
Rows: 10000 Columns: 16
── Column specification ──────────────────────────────────────────
Delimiter: ","
chr  (6): Gender, Country, Sleep_Quality, Stress_Level, Health_Issues, Occupation
dbl (10): ID, Age, Coffee_Intake, Caffeine_mg, Sleep_Hours, BMI, Heart_Rate, Physical_Activ...

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
> head(coffee_data)
# A tibble: 6 × 16
     ID   Age Gender Country Coffee_Intake Caffeine_mg Sleep_Hours Sleep_Quality   BMI
  <dbl> <dbl> <chr>  <chr>           <dbl>       <dbl>       <dbl> <chr>         <dbl>
1     1    40 Male   Germany           3.5       328.          7.5 Good           24.9
2     2    33 Male   Germany           1          94.1         6.2 Good           20
3     3    42 Male   Brazil            5.3       504.          5.9 Fair           22.7
4     4    53 Male   Germany           2.6       249.          7.3 Good           24.7
5     5    32 Female Spain             3.1       298           5.3 Fair           24.1
6     6    32 Male   Mexico            3.4       326.          6.4 Good           27
# ℹ 7 more variables: Heart_Rate <dbl>, Stress_Level <chr>, Physical_Activity_Hours <dbl>,
#   Health_Issues <chr>, Occupation <chr>, Smoking <dbl>, Alcohol_Consumption <dbl>
> high_sleep_subset <- subset(coffee_data, Sleep_Hours > 8)
> cat("Number of people with high sleep (> 8 hours):", nrow(high_sleep_subset), "\n")
Number of people with high sleep (> 8 hours): 1268
> summary(high_sleep_subset$Sleep_Hours)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  8.100   8.200   8.500   8.633   8.900  10.000
> relaxed_subset <- subset(coffee_data, Sleep_Hours > 8 & Stress_Level == "Low")
> cat("Number of relaxed people (High Sleep & Low Stress):", nrow(relaxed_subset), "\n")
Number of relaxed people (High Sleep & Low Stress): 1268
> head(relaxed_subset)
# A tibble: 6 × 16
     ID   Age Gender Country     Coffee_Intake Caffeine_mg Sleep_Hours Sleep_Quality   BMI
  <dbl> <dbl> <chr>  <chr>               <dbl>       <dbl>       <dbl> <chr>         <dbl>
1    11    29 Female Switzerland           4.5       428.          8.1 Excellent      21.5
2    33    34 Male   France                2.8       265.          8.3 Excellent      24.9
3    35    44 Female Brazil                3.8       364.          9.2 Excellent      22.7
4    53    26 Female China                 1.5       145.          9.4 Excellent      21.4
5    64    20 Male   Japan                 0           0.4         8.2 Excellent      17.2
6    69    39 Male   Canada                1.6       155           9.6 Excellent      26.5
```

```
Data
brazil_or_caffe…   1853 obs. of 16 variables
coffee_data        10000 obs. of 16 variables
Daily_Coffee_In…   10000 obs. of 16 variables
Daily_Coffee_In…   10000 obs. of 1 variable
Daily.Coffee.In…   10000 obs. of 16 variables
eu_subset_filter   983 obs. of 16 variables
female_low_caff…   859 obs. of 16 variables
high_sleep_subs…   1268 obs. of 16 variables
relaxed_subset     1268 obs. of 16 variables
young_people_fi…   2037 obs. of 16 variables
```



```
6    69    39 Male   Canada                1.6       155           9.6 Excellent      26.5
# ℹ 7 more variables: Heart_Rate <dbl>, Stress_Level <chr>, Physical_Activity_Hours <dbl>,
#   Health_Issues <chr>, Occupation <chr>, Smoking <dbl>, Alcohol_Consumption <dbl>
> brazil_or_caffeine_subset <- subset(coffee_data, Country == "Brazil" | Coffee_Intake > 4)
> cat("Number of Brazil OR High Coffee Intake:", nrow(brazil_or_caffeine_subset), "\n")
Number of Brazil OR High Coffee Intake: 1853
> head(brazil_or_caffeine_subset)
# A tibble: 6 × 16
     ID   Age Gender Country     Coffee_Intake Caffeine_mg Sleep_Hours Sleep_Quality   BMI
  <dbl> <dbl> <chr>  <chr>               <dbl>       <dbl>       <dbl> <chr>         <dbl>
1     3    42 Male   Brazil                5.3       504.          5.9 Fair           22.7
2     8    44 Female Canada                4.5       424.          5.5 Fair           15.8
3    11    29 Female Switzerland           4.5       428.          8.1 Excellent      21.5
4    12    29 Male   Netherlands           4.1       385.          6.5 Good           23.1
5    13    37 Female Italy                 4.7       442.          6.9 Good           15.4
6    19    24 Female UK                    4.3       410.          7.8 Good           22.7
# ℹ 7 more variables: Heart_Rate <dbl>, Stress_Level <chr>, Physical_Activity_Hours <dbl>,
#   Health_Issues <chr>, Occupation <chr>, Smoking <dbl>, Alcohol_Consumption <dbl>
> young_people_filter <- coffee_data |>
+   filter(Age < 25)
> young_people_filter <- coffee_data |>
+   filter(Age < 25)
> cat("Number of people under 25:", nrow(young_people_filter), "\n")
Number of people under 25: 2037
> summary(young_people_filter$Age)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  18.00   18.00   19.00   20.07   22.00   24.00
> female_low_caffeine_filter <- coffee_data |>
+   filter(Gender == "Female", Caffeine_mg < 100)
> female_low_caffeine_filter <- coffee_data |>
+   filter(Gender == "Female", Caffeine_mg < 100)
> cat("Number of Females with Low Caffeine:", nrow(female_low_caffeine_filter), "\n")
Number of Females with Low Caffeine: 859
> head(female_low_caffeine_filter)
# A tibble: 6 × 16
     ID   Age Gender Country     Coffee_Intake Caffeine_mg Sleep_Hours Sleep_Quality   BMI
  <dbl> <dbl> <chr>  <chr>               <dbl>       <dbl>       <dbl> <chr>         <dbl>
1    21    52 Female Italy                 0.7        65.5         6.4 Good           28.2
2    23    35 Female Belgium               0           0           5.5 Fair           23.4
3    26    36 Female Mexico                0.4        34.7         6.1 Good           15
```

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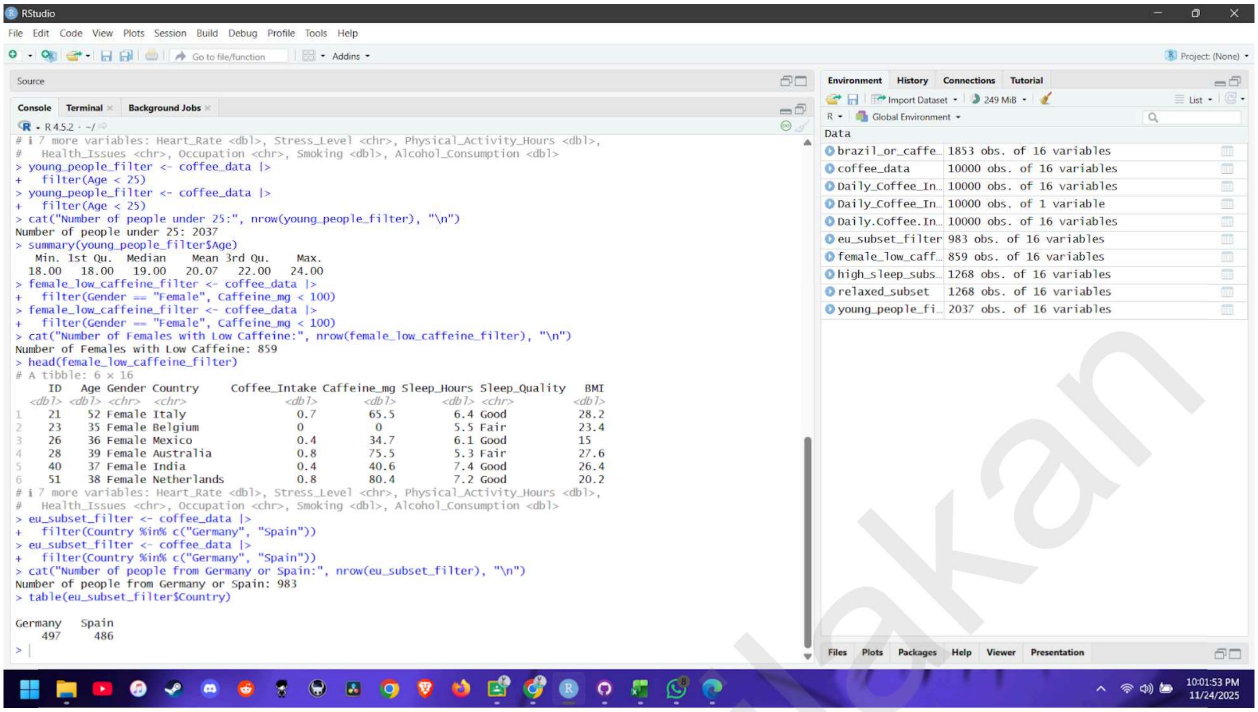
S097


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Addins
Project: (None)
Source
Console
Terminal
Background Jobs
R 4.5.2 · ~/
i 7 more variables: Heart_Rate <dbl>, Stress_Level <chr>, Physical_Activity_Hours <dbl>,
Health_Issues <chr>, Occupation <chr>, Smoking <dbl>, Alcohol_Consumption <dbl>
> young_people_filter <- coffee_data |>
+ filter(Age < 25)
> young_people_filter <- coffee_data |>
+ filter(Age < 25)
> cat("Number of people under 25:", nrow(young_people_filter), "\n")
Number of people under 25: 2037
> summary(young_people_filter$Age)
Min. 1st Qu. Median Mean 3rd Qu. Max.
18.00 18.00 19.00 20.07 22.00 24.00
> female_low_caffeine_filter <- coffee_data |>
+ filter(Gender == "Female", Caffeine_mg < 100)
> female_low_caffeine_filter <- coffee_data |>
+ filter(Gender == "Female", Caffeine_mg < 100)
> cat("Number of Females with Low Caffeine:", nrow(female_low_caffeine_filter), "\n")
Number of Females with Low Caffeine: 859
> head(female_low_caffeine_filter)
A tibble: 6 × 16
ID Age Gender Country Coffee_Intake Caffeine_mg Sleep_Hours Sleep_Quality BMI
<dbl> <dbl> <chr> <chr> <dbl> <dbl> <dbl> <chr> <dbl>
1 21 52 Female Italy 0.7 65.5 6.4 Good 28.2
2 23 35 Female Belgium 0 0 5.5 Fair 23.4
3 26 36 Female Mexico 0.4 34.7 6.1 Good 15
4 28 39 Female Australia 0.8 75.5 5.3 Fair 27.6
5 40 37 Female India 0.4 40.6 7.4 Good 26.4
6 51 38 Female Netherlands 0.8 80.4 7.2 Good 20.2
i 7 more variables: Heart_Rate <dbl>, Stress_Level <chr>, Physical_Activity_Hours <dbl>,
Health_Issues <chr>, Occupation <chr>, Smoking <dbl>, Alcohol_Consumption <dbl>
> eu_subset_filter <- coffee_data |>
+ filter(Country %in% c("Germany", "Spain"))
> eu_subset_filter <- coffee_data |>
+ filter(Country %in% c("Germany", "Spain"))
> cat("Number of people from Germany or Spain:", nrow(eu_subset_filter), "\n")
Number of people from Germany or Spain: 983
> table(eu_subset_filter$Country)
Germany Spain
497 486
>
Environment History Connections Tutorial
Import Dataset
249 MiB
List
R
Global Environment
Data
brazil_or_caffe… 1853 obs. of 16 variables
coffee_data 10000 obs. of 16 variables
Daily_Coffee_In… 10000 obs. of 16 variables
Daily_Coffee_In… 10000 obs. of 1 variable
Daily.Coffee.In… 10000 obs. of 16 variables
eu_subset_filter 983 obs. of 16 variables
female_low_caff… 859 obs. of 16 variables
high_sleep_subs… 1268 obs. of 16 variables
relaxed_subset 1268 obs. of 16 variables
young_people_fi… 2037 obs. of 16 variables
Files Plots Packages Help Viewer Presentation
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